

Shailesh Dudala

SENIOR APPLIED AI ENGINEER | HEALTHCARE & INSURANCE

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PROFILE

Applied AI engineer with more than seven years across healthcare, insurance, document intelligence, predictive ML, and MLOps. Designs the contracts, evaluation, human review, and observability that turn uncertain model output into accountable workflows.

CORE

Agentic workflows | Document intelligence, OCR, and RAG | Healthcare interoperability (FHIR/HL7) | Predictive ML | Human-in-the-loop systems | Evaluation and observability | Local/private AI | ML platforms and MLOps

PROFESSIONAL EXPERIENCE

Applied AI Engineer Consultant | MetLife via Bizintex

2026 - Present

- Designs claims and document workflows with typed extraction, validation gates, reviewer fallbacks, and trace telemetry.
- Supported approximately 90% lower handling effort and approximately 50% shorter time-to-claim-payable in a measured claims workstream.

Lead DS/ML Engineer Consultant | Inland Empire Health Plan via Infowave

2023 - 2025

- Led local document intelligence, quality-measure evidence, predictive analytics, fraud/waste review, and MLOps.
- Cleared a 7,000-case backlog with approximately 90% lower review time inside the local data boundary.
- Improved automated quality-measure closures by 20% and reduced transportation waste by 18% in their respective workflows.

Lead Data Scientist / Product | Hexplora

2020 - 2023

- Helped build and scale a 0-to-1 analytics product across nine healthcare programs, spanning risk models, reusable delivery, dashboards, and care workflows.
- Supported \$500K in new revenue, approximately \$3M in broader client P4P impact, and approximately 50% faster model deployment cycles.

Data Science and Biomedical Informatics Roles | CommonSpirit Health; Health New England; University of Chicago; AbbVie

Earlier foundations

- Worked across hospital analytics, provider data, biomedical research, clinical sensors, genomics, and public-health modeling.
- Delivered analytics spanning 142 hospitals and validation covering 100,000+ provider records; reduced manual provider review by 60% in a measured workflow.

SELECTED SYSTEMS

- Claims intelligence - typed extraction, deterministic validation, exception routing, human review, and audit telemetry.
- On-prem document intelligence - local parsing, retrieval, page citations, and explicit fallback states.
- Healthcare analytics platform - risk models, reusable delivery, reporting, and care workflows across nine programs.
- LLM Steering - public local-first activation-steering workbench with tests, experiment manifests, and documented limits.

EDUCATION + CREDENTIALS

M.S., Biomedical Informatics - University of Chicago, 2019

B.Tech., Computer Science and Engineering - SRM University, 2018

Yale Summer School, Public Health Modeling, 2019

AWS ML Engineer - Associate | Azure AI Fundamentals | Power BI Data Analyst

RECOGNITION

Team recipient - Global HL7 AI Challenge, Transformative Impact in Healthcare Award (2025), Let's Talk Doc

PUBLICATION BOUNDARY: Professional systems are described without PHI, PII, employer-confidential artifacts, proprietary workflow rules, credentials, or private production details.